

Computer Vision Fundamentals: Image Preprocessing

Before running detection models, images are typically converted to grayscale to reduce computational load, then a Gaussian blur is applied to suppress high-frequency noise. Common kernel sizes are 3x3 or 5x5. Histogram equalization can improve contrast in poorly lit images and is especially useful for industrial inspection cameras with inconsistent lighting.

Computer Vision Fundamentals: Edge Detection and Contours

The Canny edge detector is the standard method for edge detection, using two threshold values to classify strong and weak edges. After edge detection, contours can be extracted using `findContours`, which returns a hierarchy of boundary curves. Contour area and perimeter are commonly used to filter out noise blobs from real objects of interest.

Computer Vision Fundamentals: Object Detection Pipelines

Modern object detection pipelines (YOLO, Faster R-CNN) output bounding boxes with class labels and confidence scores. A confidence threshold of 0.5 is a common default, but should be tuned per use case. Non-maximum suppression (NMS) removes duplicate overlapping boxes for the same object, typically using an IoU threshold of 0.4 to 0.5.